Assignment 01 - Circuit Simulation using NGSPICE

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September 01, 2026

1 Objective

To analyze wideband RC voltage divider in both the time and frequency domain using NGSPICE. The circuit is evaluated using three values of compensation capacitor C1 while keeping R1, R2 and C2 fixed.

The main idea of the experiemnt is that a voltage divider inteaded for a wide frequency rangne must preserve approximately the same attenuation for both low-frequency and high frequency components.

2 Schematic

The upper branch consists of R1 in parallel with C1, while the lower branch consists of R2 in parallel with C2. The output voltage is measured at the junction of the two branches.

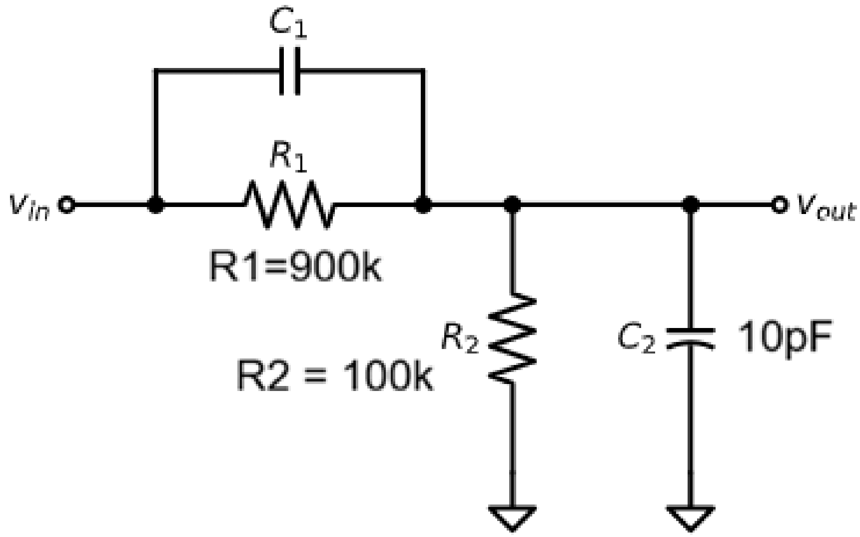

Figure 1: Wideband RC Voltage Divider Circuit

3 Theoretical Analysis

3.1 Low-frequency behavior

At DC or at low frequencies, the capacitors behave as open circuit. The circuit therefore reduces to an ordinary resistive voltage divider:

$$\frac{Vout}{Vin} = \frac{R_2}{R_1 + R_2}$$

Substituting the resistor values,

$$\frac{Vout}{Vin} = \frac{100k\Omega}{100k\Omega + 900k\Omega} = 0.1$$

Thus, the circuit attenuates the Vin to 0.1. For an input of +1V, the low-frequency output is equal to +0.1V, and for an input of -1V, the output is equal to -0.1V.

In decibels, the magnitude is equal to:

$$20log_{10} = -20dB$$

3.2 High-frequency behavior

As frequency increases, the capacitor impedance decreases according to:

$$X_C = \frac{1}{2\pi fC}$$

At very high frequencies, the limiting voltage ratio approaches up to:

$$\frac{Vout}{Vin} = \frac{C_1}{C_1 + C_2}$$

For the divider to have the same attenuation at low and high frequencies, the resistive and capacitive division ratios must be equal:

$$\frac{R_2}{R_1 + R_2} = \frac{C_1}{C_1 + C_2}$$

Rearranging gives the compensation condition:

$$R_1C_1 = R_2C_2$$

3.3 Finding Compensation Value

To solve the required value of C1,

$$C_1 = \frac{R_2C_2}{R_1}$$

Using the given component values,

$$C_1 = \frac{(100k\Omega)(10pF)}{(900k\Omega)} = 1.111pF$$

Therefore, C1 = 1.111pF is correctly compensated case. To show the result for both undercompensated and overcompensated case:

C1	Result
1.1111pF	Correctly Compensated
1.6pF	Overcompensated
0.7pF	Undercompensated

Table 1: C1 Compensation Values

4 NGSPICE Simulation Setup

The COLAB notebook defines the three version of the circuit in one NGSPICE netlist. The input source is a square wave that switches between -1V and +1V with a period of 10us. Therefore, the square-wave frequency is:

$$f = \frac{1}{10\mu s}$$

The transient analysis uses (tran 0.01 u 0.3m) which observes the circuit for approximately 30

$$\mu$$

s. The AC analysis uses (ac dec 10 1 1G) which sweeps the frequency from 1Hz to 1GHz with 10 points per decade. The AC source magnitude is 1V, so the plotted output magnitude directly represents the circuit gain.

5 Results and Discussion

5.1 Transient Response

Figure 02 shows the transient response of the RC Voltage Divider. For C1 = 1.111pF, the output changes cleanly between -0.1 V and +0.1V. This occurs since we have solved that the time constants will be matched.

As a result, the fast components of the square wave experience approximately the same attenuation as the slow components.

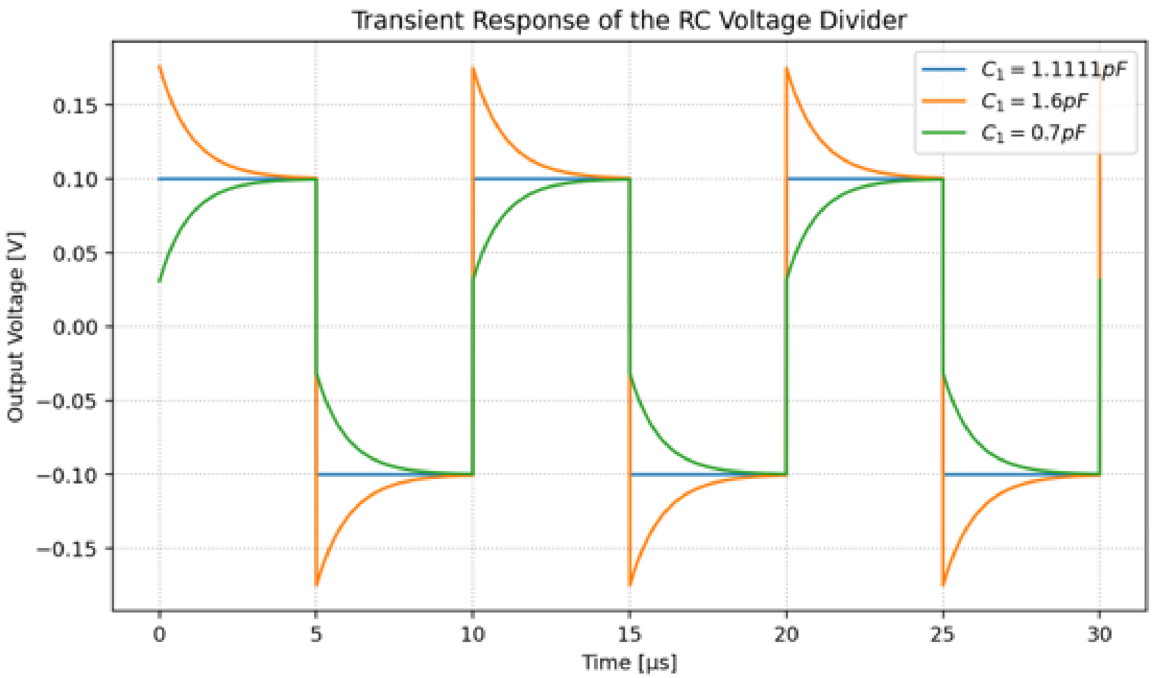

Figure 2: Transient Response of the RC Voltage Divider

For C1 = 1.6pF, the output initially exceeds its final value after each transition. This is overshoot, and it indicates overcompensation. The high-frequency attenuation is smaller than the low-frequency attenuation, so the fast components of the square-wave edge are emphasized.

For C1 = 0.7 pF, the output initially changes by a smaller amount and then gradually settles toward ±0.1 V. This is an undercompensated response. High-frequency components are attenuated more strongly, causing the square-wave edges to become rounded.

5.2 Magnitude Frequency Response

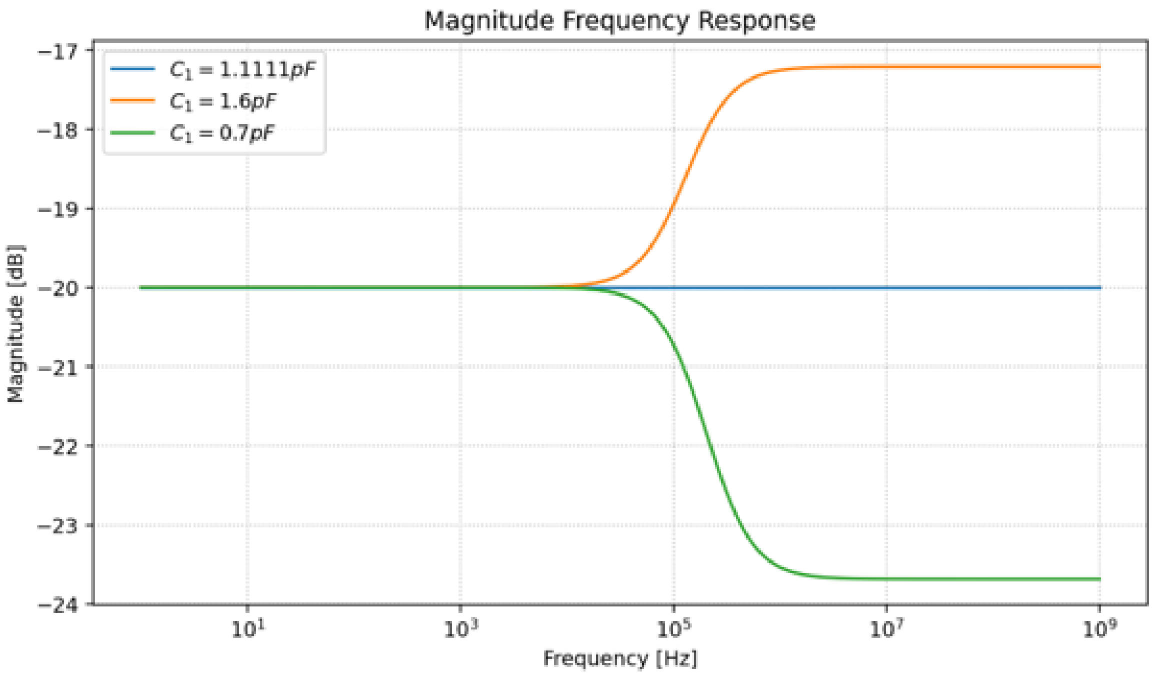

Figure 3: Magnitude Frequency Response

At low frequency, all three cases are controlled mainly by the resistive divider so each begins at -20dB.

For the correctly compensated value, C1 = 1.1111pF:

$$\frac{C_1}{C_1 + C_2} = 0.1$$

so the high frequency gain is also -20dB. The response therefore remains flat over the entire frequency range.

For C1 = 1.6pF,

$$\frac{C_1}{C_1 + C_2} = \frac{1.6}{1.6 + 10} = 0.1379$$

which corresponds to approximately

$$20log(0.1379) = -17.2dB$$

The high-frequency magnitude is therefore greater than the low-frequency magnitude, explaining the upward bend of the response and the overshoot in the transient waveform.

For C2 = 0.7pF,

$$\frac{C_1}{C_1 + C_2} = \frac{0.7}{0.7 + 10} = 0.0654$$

which corresponds to approximately

$$20log(0.0664) = -23.7dB$$

The high-frequency magnitude is therefore lower than the low-frequency magnitude, which explains the rounded transient response.

5.3 Phase Frequency Response

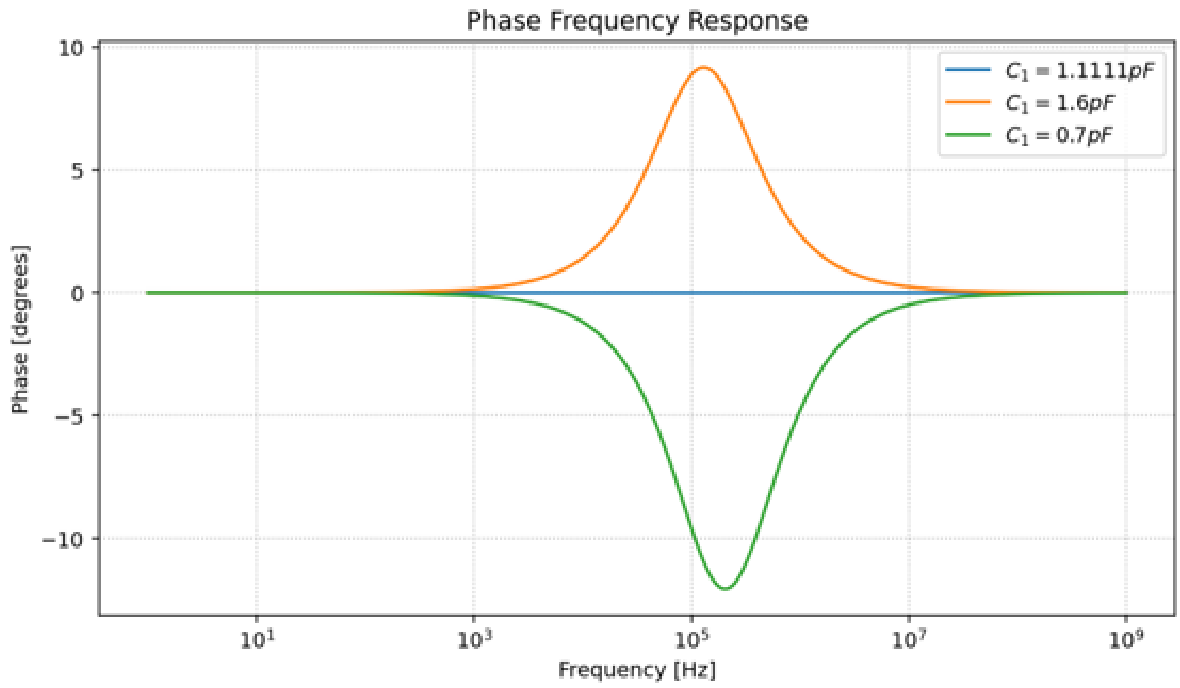

Figure 4: Phase Frequency Response

The correctly compensated case remains close to because the frequency-dependent effects of the two RC sections balance each other.

The case shows a positive phase shift over the transition region. This is consistent with a **phase-lead** behavior and with the magnitude increasing at higher frequencies.

The case shows a negative phase shift over the transition region. This is consistent with **phase lag** and with stronger attenuation of high-frequency components.

At very low and very high frequencies, the phase of all three cases approaches because the transfer ratio approaches a nearly real constant value at each frequency extreme.

6 Question and Answer

6.1 Q1: Where can this circuit be used?

This circuit can be used as a frequency-compensated voltage divider. The resistor provides the desired low-frequency attenuation, while the capacitors compensate for the frequency-dependent input capacitance. When the divider is correctly compensated, fast signals can be attenuated without significantly changing their waveform shape.

6.2 Q2: Why do the magnitude and phase frequency response plots behave this way?

The behavior is determined by the relationship between the two RC time constants. When C1=1.1111pF, the condition $R_1C_1 \approx R_2C_2$ is satisfied. The low-frequency and high-frequency attenuation are therefore approximately equal, producing a nearly constant magnitude of -20dB and a phase close to 0°.

When C1 is increased to 1.6 pF, the divider is overcompensated. The high-frequency gain becomes larger than the low-frequency gain, so the magnitude response rises, and the phase becomes positive over the transition region. This also produces overshoot in the transient response.

When C1 is decreased to 0.7pF, the divider is undercompensated. High-frequency components are attenuated more strongly, so the magnitude response falls and the phase becomes negative over the transition region. This causes the slow or rounded edges observed in the transient response.

7 Conclusion

This activity demonstrates that a resistive divider alone does not necessarily preserve its attenuation when signal frequency becomes high enough for circuit capacitances to matter. By adding the compensation capacitor C1 and satisfying

$$R_1C_1 = R_2C_2$$

the divider can maintain approximately the same attenuation across a much wider frequency range. For the given values, the correct compensation capacitor is C1=1.1111pF. A larger value causes overcompensation and overshoot, while a smaller value causes undercompensation and rounded transitions. The experiment therefore connects time-domain waveform distortion with the corresponding magnitude and phase behavior in the frequency domain.